Population, Sample and Sampling Methods

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1. Population and Sample – Real-World Examples

A **population** means the complete group of people or items that we want to study. A **sample** is a smaller group selected from that population for analysis.

Examples Where I Am Part of a Population

1. **Instagram:** If Instagram studies all users who watch Reels in India, I am part of the **population** if I watch Reels.
2. **Zomato:** If Zomato studies all customers who order food in Ahmedabad, I am part of the **population** if I order food through Zomato.
3. **Flipkart:** If Flipkart studies all customers who purchase mobile phones, I am part of the **population** if I have purchased a mobile phone from Flipkart.

Examples Where I Am Part of a Sample

1. **Instagram:** If Instagram selects 1,000 users from millions of users to ask about a new feature and I am selected, I am part of the **sample**.
2. **Zomato:** If Zomato randomly selects 50 customers from Ahmedabad for a feedback survey and I am selected, I am part of the **sample**.
3. **Flipkart:** If Flipkart selects 500 customers to test a new shopping feature and I am selected, I am part of the **sample**.

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2. Random Sampling Plan for Zomato

Situation:

Zomato wants to find the **average rating of restaurants in Ahmedabad** and needs to select 10 restaurants for a survey.

Sampling Plan:

1. First, Zomato prepares a complete list of restaurants in Ahmedabad.
2. Each restaurant is given a unique number.
3. From this list, **10 restaurants are selected randomly**.
4. The selected restaurants are contacted and their ratings are collected.
5. The average rating of these 10 restaurants can then be calculated.

Why is Random Sampling Important?

Random sampling gives every restaurant an **equal chance of being selected**. It reduces personal bias and makes the sample more representative of restaurants across Ahmedabad.

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3. Flipkart Survey Sampling

Situation:

Flipkart wants to survey users about a new feature but sends the survey **only to users who made a purchase last week**.

Type of Sampling:

This is a form of **convenience sampling** because Flipkart is selecting users who are easily available or meet a specific convenient condition.

Possible Source of Bias:

Users who did not make a purchase last week have **no chance of being selected**. Their opinions about the new feature may be different from recent buyers.

Therefore, the survey may not represent the opinions of **all Flipkart users**.

Conclusion:

The sample can be biased because the survey only includes **recent purchasers** instead of users from the entire population.

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4. Customer Feedback Survey – Instagram

App Selected: Instagram

Purpose of Survey:

To understand how users feel about the Instagram Reels feature.

Sample Size:

20 users

Survey Questions:

1. How often do you watch Instagram Reels?
    - Daily
    - Several times a week
    - Occasionally
    - Rarely
2. How satisfied are you with Instagram Reels?
    - Very Satisfied
    - Satisfied
    - Neutral
    - Dissatisfied
    - Very Dissatisfied
3. How easy is it to discover new content through Reels?
    - Very Easy
    - Easy
    - Neutral
    - Difficult
4. What type of Reels do you watch most?
    - Music
    - Comedy
    - Sports
    - Education
    - Other
5. What improvement would you like to see in Reels?

Sampling Method: Stratified Sampling

I would use **stratified sampling** to select the 20 users.

First, users can be divided into groups based on their **age group or Reels usage frequency**. Then, users would be randomly selected from each group.

Why Choose Stratified Sampling?

Different groups of Instagram users may have different opinions about Reels. Stratified sampling ensures that **different types of users are represented** in the survey instead of selecting all 20 users from only one group.